LBEADS-NET

Algorithm Unrolling for Learned Baseline Estimation and Denoising in Chromatographic Signals

MS Thesis Defense

Saad Zubairi

Advisor: Professor Ivan Selesnick

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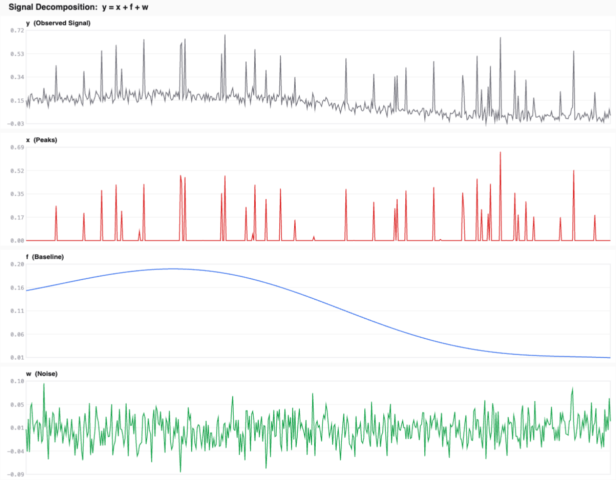
New York University
Tandon School of Engineering

Act I: The Problem

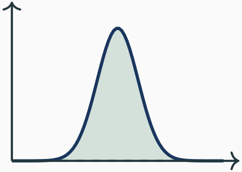
What is a Chromatogram?



The observed signal is a mixture of three components: $y = x + f + w$

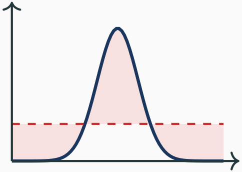
Why Baseline Correction Matters

Correct Baseline



Accurate area
⇒ Correct conc. ✓

Wrong Baseline



Wrong area
⇒ Wrong conc. ✗

In pharmaceutical QC, even 2% baseline error can exceed regulatory limits.

The Classical Solution: BEADS

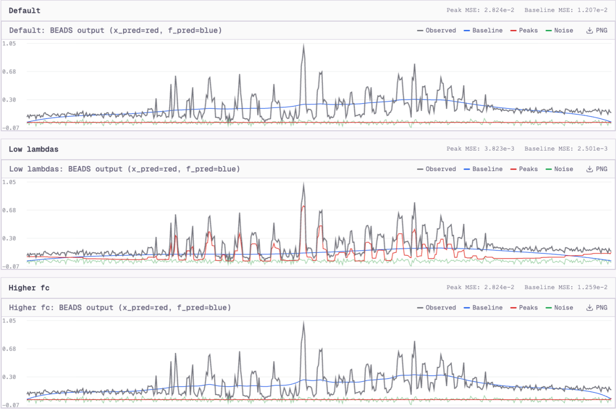
Baseline **E**stimation **A**nd **D**enoising using **S**parsity (Ning et al., 2014)

$$J(\mathrm{x}) = \frac{1}{2}\|H(\mathrm{y}-\mathrm{x})\|_2^2 + \lambda_0 \sum_n \theta(x_n) + \lambda_1 \sum_n \phi(\Delta x_n) + \lambda_2 \sum_n \phi(\Delta^2 x_n)$$

1. **Data fidelity** — high-pass filter H separates baseline from peaks
2. **Asymmetric sparsity penalty** θ — peaks are sparse and non-negative
3. **First-order smoothness** — penalizes peak roughness
4. **Second-order smoothness** — penalizes peak curvature

6 parameters to tune: $\lambda_0,\ \lambda_1,\ \lambda_2,\ r,\ f_c,\ d$

The Tuning Problem



No single parameter configuration generalizes across instruments, analytes, or conditions.

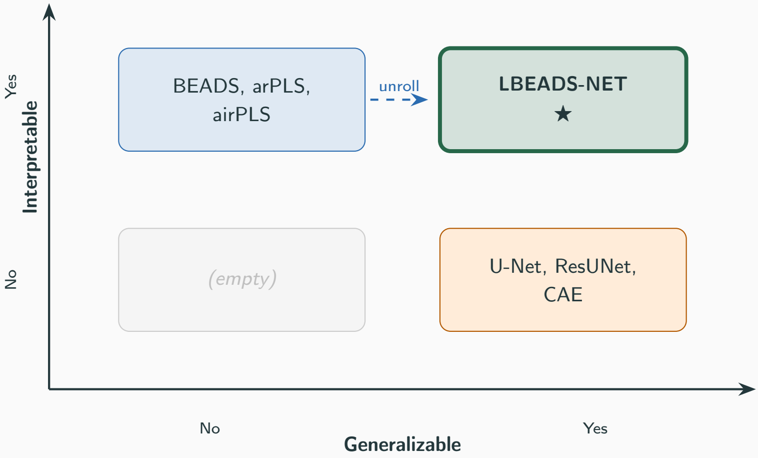
The Deep Learning Alternative

- **Convolutional autoencoders** (Kensert et al., 2021) — 190K synthetic chromatograms
- **ResUNet** (Chen et al., 2022) — 2.35M parameters, 210K spectra
- **DIRAS+** (Aradhye et al., 2025) — hybrid CNN + classical solver

These methods generalize without per-signal tuning. . .

. . . but they are **black boxes** with millions of opaque parameters.

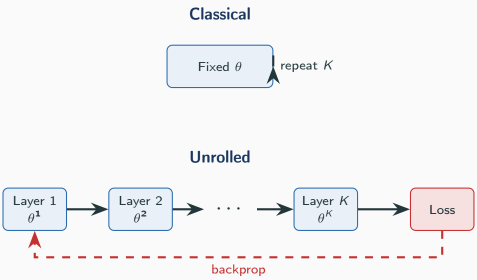
The Gap



Yes
Interpretable
No
BEADS, arPLS,
airPLS
unroll
LBEADS-NET
★
(empty)
U-Net, ResUNet,
CAE
No
Generalizable
Yes

Our Approach: Algorithm Unrolling

- Map each **iteration** of an optimization algorithm to one **neural network layer**
- Algorithm parameters $\rightarrow$ **learnable** network parameters
- K layers $=$ K iterations, trained **end-to-end**



LBEADS-NET:
20 layers $\times$ 5 parameters $=$ **100 learnable scalars**

Act II: The Attempt

From BEADS to BEADSLayer

Classical BEADS (one iteration)

1. Compute reweighting matrices Λ, Γ
2. Assemble system matrix M
3. Solve banded linear system
   $(B^\top B + A^\top M A)\,\mathrm{z} = \mathrm{d}$
4. Update $\mathrm{x}^{(k+1)} = A\mathrm{z}$

$\xrightarrow{\textit{reform.}}$

ISTA Layer (one layer)

1. Gradient step:
   $$\mathrm{x}_{\mathrm{up}} = \mathrm{x}^{k-1} + \eta\, \mathrm{g}_{\mathrm{data}} - \eta(\nabla\mathcal{R}_1 + \nabla\mathcal{R}_2)$$
2. Proximal step:
   $$\mathrm{x}^{k} = \mathcal{S}_{\mathrm{asym}}(\mathrm{x}_{\mathrm{up}};\ \lambda_0\eta,\ r)$$

Replace exact linear system solve with **gradient descent + proximal operator**.

The ISTA Layer

Gradient step:

$$\mathrm{x}_{\mathrm{update}} = \mathrm{x}^{k-1} + \eta^k \, \mathrm{g}_{\mathrm{data}} - \eta^k (\nabla_{\mathrm{x}} \mathcal{R}_1 + \nabla_{\mathrm{x}} \mathcal{R}_2)$$

Proximal step (asymmetric soft-thresholding):

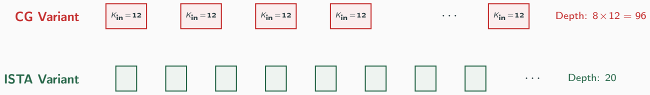
$$[\mathrm{x}^k]_i = \begin{cases} [\mathrm{x}_{\mathrm{update}}]_i - \lambda_0^k \eta^k & \text{if } [\mathrm{x}_{\mathrm{update}}]_i > \lambda_0^k \eta^k \\ [\mathrm{x}_{\mathrm{update}}]_i + \lambda_0^k \eta^k r^k & \text{if } [\mathrm{x}_{\mathrm{update}}]_i < -\lambda_0^k \eta^k r^k \\ 0 & \text{otherwise} \end{cases}$$

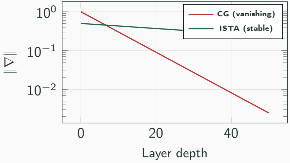
$\lambda_0 \eta$ — **effective threshold**: controls sparsity level

r^k — **asymmetry ratio**: suppresses negative artifacts

Warning: The proximal operator shrinks **ALL** non-zero values $\rightarrow$ source of baseline leakage

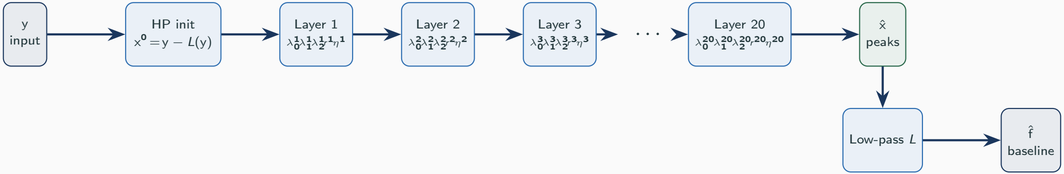
Why Not Conjugate Gradient?





Simpler layers $\times$ more layers $>$ Complex layers $\times$ fewer layers

Full Architecture



Signal length $N = 4096$ | $K = 20$ layers | 5 params/layer | **100 total learnable scalars**

The Development Timeline

1.0	1.1	1.2	1.3	1.4	1.5
Classical BEADS	**MSE Only**	**+Sparsity**	**+Baseline Supervision**	**+Orthogonality (v5)**	**Full v7**
– terms	1 terms	4 terms	6 terms	8 terms	12 terms
Fixed params	Peak MSE	+ ℓ_1 sparsity + TV penalty + Non-negativity	+ Masked baseline reconstruction + Baseline smoothness	+ Peak–baseline orthogonality + 3rd-deriv. penalty	+ Asymmetric BL + Leakage penalty + Envelope + Freq. separation

Progressive Development: Each experiment adds one key idea

Three-Stage Curriculum Training

Stage A
Peak Reconstruction
Peak MSE only
1 active term

Learn basic separation

→ *add anti-leakage* →

Stage B
Leakage Mitigation
+ Sparsity, TV, non-neg.
+ Baseline supervision
+ Orthogonality

10 active terms

Fix leakage pathways

→ *activate remaining* →

Stage C
Full Refinement
All terms active
Rebalanced weights
12 active terms

Fine-tune decomposition

Why curriculum? Activating 12 loss terms from scratch overwhelms 100 parameters.
Stage A provides a stable initialization before anti-leakage penalties introduce competing gradients.

Thesis model (v5): Stage A → Stage B only (8 terms). **v7:** All three stages (12 terms) — performed worse.

Exp 1.0: Classical BEADS (Baseline)

- Apply classical BEADS with published default parameters
- Parameters designed for unnormalized data (amplitudes $\sim$100s)
- Applied to normalized $[0, 1]$ test signals

Metric	Value
Area Error	**99.99%**
Peak Correlation	0.746
Baseline Leakage Index	0.837

Almost all peak content treated as baseline.

Takeaway: Default parameters catastrophically fail on normalized data — **the tuning problem in its most acute form**.

Exp 1.1: Naive Unrolling (MSE Only)

Train with reconstruction loss only: $\alpha_{\mathsf{mse}} = 1.0$, all others $= 0$.

Metric	**Classical**	**Exp 1.1**
Peak MSE	21.27	**9.79 ✓**
Peak Corr.	0.746	0.748
Area Error	99.99%	**49.47% ✓**
BLI	0.837	**0.593 ✓**
Neg. Fraction	0.000	0.295 ✗

Algorithm unrolling works — but the architecture has an inherent sparsity bias.

Exp 1.2: + Sparsity Priors

Add ℓ_1 sparsity, total variation, non-negativity penalties.

Improved:

Peak Corr: 0.748 $\rightarrow$ **0.785** ✓

Neg. Fraction: 0.295 $\rightarrow$ **0.216** ✓

Unchanged:

BLI: 0.593 $\rightarrow$ 0.597 ✗

Area Error: 49.47% $\rightarrow$ 49.51% ✗

Sparsity priors improve peak *shape* but do **NOT** address the leakage mechanism.

Exp 1.3: + Baseline Supervision

Add masked baseline reconstruction + baseline smoothness.

Metric	Exp 1.2	Exp 1.3
Peak MSE	9.79	**9.30**
Peak Corr.	0.785	**0.793**
Baseline MSE	—	7.48

The single most impactful addition.

Direct gradient signal about baseline quality provides information that pure peak reconstruction cannot.

Only 6 active loss terms.

Exp 1.4: + Orthogonality (v5 — Thesis Model)

Add peak–baseline orthogonality + baseline third-derivative penalty.

Metric	Exp 1.3	Exp 1.4
Baseline MSE	7.48	**7.40 ✓**
BLI	0.596	**0.592 ✓**
Peak Corr.	0.793	0.785 (mild decrease)

Enforcing mutual exclusivity between peaks and baseline improves baseline quality at a mild cost to peak shape.

This is the thesis model configuration.

Exp 1.5: Full Anti-Leakage Battery (v7)

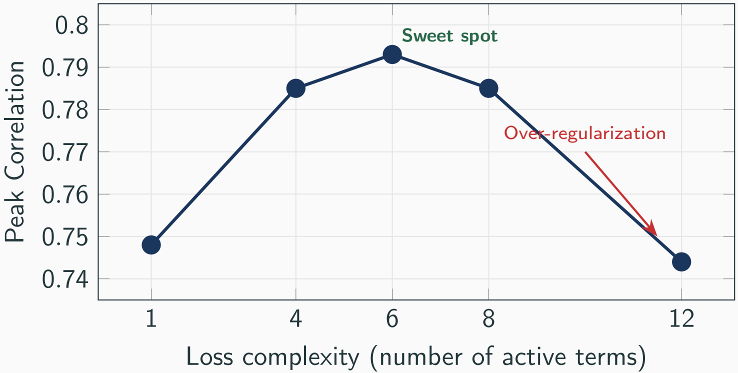
Activate **ALL 12 loss terms**: asymmetric baseline, leakage penalty, envelope constraint, frequency separation.

Metric	Exp 1.4	Exp 1.5
Peak Corr.	0.785	**0.744 ✗**
Peak MSE	9.36	**10.01 ✗**
Neg. Fraction	0.211	**0.305 ✗**
BLI	0.592	**0.611 ✗**

More loss terms $\neq$ better performance.

12 competing objectives overwhelm 100 parameters in 300 gradient steps.

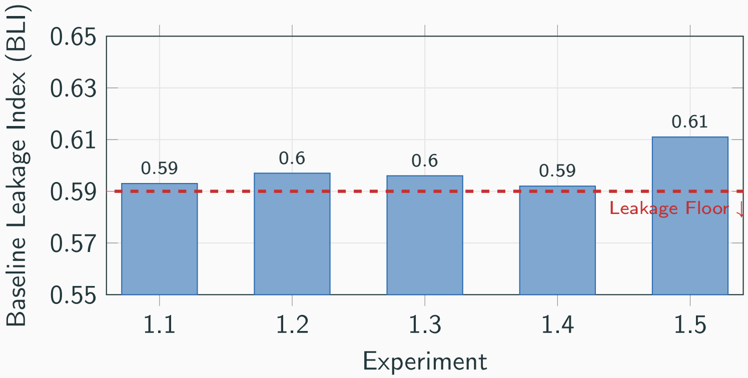
The Over-Regularization Lesson



The relationship between loss complexity and model quality is **non-monotonic**.

This motivated the 3-stage curriculum training strategy.

The BLI Plateau



BLI varies by only **0.005** across Experiments 1.1–1.4.

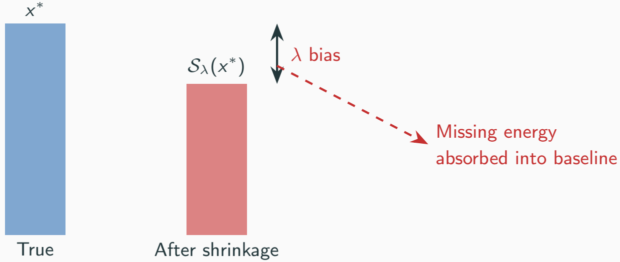
Loss engineering cannot overcome the architectural inductive bias.

Act III: The Discovery

Why Leakage is Fundamental

Soft-thresholding:

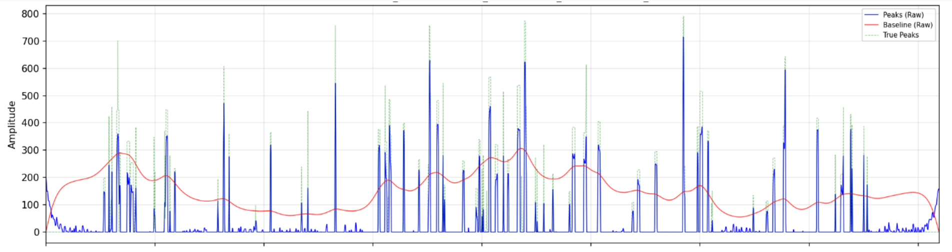
$$\mathcal{S}_\lambda(x) = \mathrm{sign}(x) \cdot \max(|x| - \lambda,\ 0)$$



For **any** $\lambda > 0$, every non-zero signal value is shrunk by exactly λ.

In dense-peak regions where the sparsity assumption fails, this shrinkage accumulates across all peak samples.

Leakage in Action

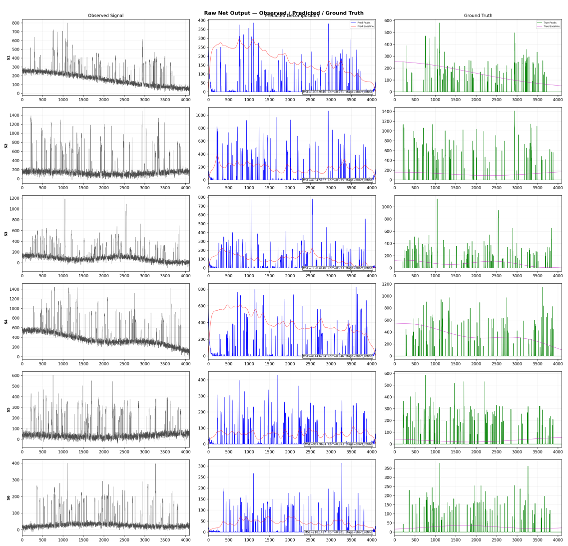


Soft-threshold shrinks **all** peaks by λ → Dense regions: shrinkage $\times$ many samples $=$ large energy deficit → Low-pass filter absorbs missing energy $\rightarrow$ **baseline bulges up**

Green $=$ true peaks Blue $=$ predicted peaks (shrunk) Red $=$ baseline (inflated)

Where peaks are dense, the baseline **absorbs the energy** that soft-thresholding removes.

Raw Network Output: Observed / Predicted / Ground Truth



Three Root Causes of Baseline Leakage

Sparsity Assumption Violation

Dense peaks overlap $\rightarrow$ signal is NOT sparse

ℓ_1 over-shrinks $\rightarrow$ energy leaks to baseline

Fixed Cutoff Frequency

$f_c = 0.006$ for ALL signals

Broad peaks have energy below $f_c \rightarrow$ attributed to baseline

Spatially Uniform Regularization

Same λ everywhere

Dense regions need weak λ, sparse regions need strong λ

All three compound in **dense-peak regions** — where accurate baseline correction is most needed.

Emergent Parameter Specialization

Parameter	Layer 0	Layer 3	Layer 5
λ_0 (sparsity)	0.481	1.198	2.000*
r (asymmetry)	3.36	2.00*	9.67
η (step size)	0.248	0.191	0.243

- **Early layers**: gentle, conservative separation (low λ_0, moderate r)
- **Late layers**: aggressive refinement (high λ_0, strong asymmetry)
- **Step size**: determined by filter spectral properties, not layer position

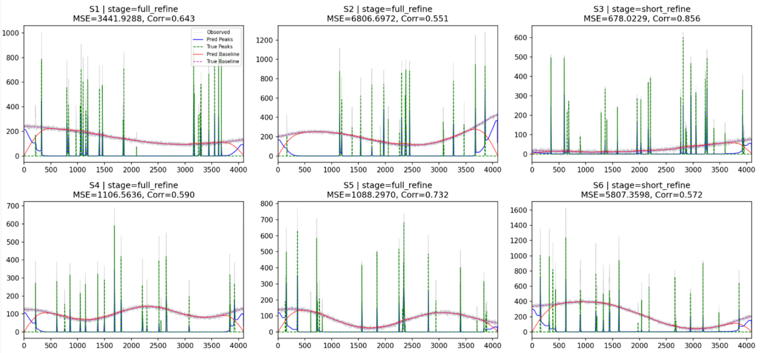
This specialization **emerges from training** — all layers initialized identically.

What Actually Worked

- ✓ Algorithm unrolling eliminates per-signal tuning
  ($2\times$ MSE improvement over default BEADS)
- ✓ Masked baseline supervision is the single most impactful loss term
- ▲ Baseline leakage persists at BLI ≈ 0.59 — a fundamental ℓ_1 limit
- ✗ More loss terms $\neq$ better: over-regularization degrades performance
- ✓ Learned parameters exhibit emergent layer specialization

The negative results (3, 4) are **as important as the positive ones**.

Fixing Leakage: The Peak Trade-Off



Baseline improves with anti-leakage terms, but **peaks degrade** — the same penalties over-shrink peak amplitudes.

Limitations

- Quantitative evaluation on **synthetic data** — real GC chromatographs tested qualitatively (visually plausible baselines, but no ground truth for metrics)
- No comparison with arPLS, airPLS, or deep learning baselines
- Dense $O(N^2)$ matrices — 128 MB per buffer at $N = 4096$
- Fixed cutoff frequency f_c (not learnable due to SciPy–PyTorch boundary)
- $\sim$50% area error on synthetic data — not deployment-ready for quantitative analysis

> This is a **methodological exploration** — but the architecture is promising for real-world deployment.

Future Directions

1. **Group sparsity ($\ell_{2,1}$ penalty)**
   Replace element-wise ℓ_1 with block soft-thresholding.
   Preserves within-group amplitudes $\rightarrow$ directly addresses leakage.
2. **Banded matrix implementation in PyTorch**
   $O(N^2) \rightarrow O(N \cdot b)$. Enables $N = 100{,}000+$ signals natively on GPU.
3. **Real chromatographic validation**
   Spike-and-recovery experiments, consensus labeling with expert chromatographers.
   The critical missing piece for deployment.
4. **Edge deployment on STM32 microcontrollers**
   Only **100 scalar parameters** + banded ops $\rightarrow$ tiny memory footprint.
   At $N=4096$ with banded matrices: $\sim$200 KB RAM (signal buffers + filter bands).
   Feasible on STM32H7-class MCUs (1 MB+ SRAM, Cortex-M7 @ 480 MHz).
   Enables real-time baseline correction on portable instruments.

Thank You

Algorithm unrolling bridges interpretability and generalization —
and reveals the fundamental limits of sparsity-based signal decomposition.

Advisor: Professor Ivan Selesnick

Saad Zubairi — NYU Tandon — May 2026

Questions?